

Bike Demand Forecasting Hackathon

Introduction to Machine Learning (67577)

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1 Data Exploration and Understanding

1.1 Dataset Overview

The training data (`train_set.csv`) contains 1,605,208 ride-level logs from three cities spanning January 1, 2025 to February 28, 2025 (58 days). Each row represents a single bike ride with 34 features covering temporal, weather, station metadata, and POI information.

1.2 City Distribution

- **City 1:** 1,142,317 rides (71.2%), 807 stations
- **City 2:** 460,642 rides (28.7%), 783 stations
- **City 3:** 2,249 rides (0.1%), 211 stations

City 1 is the dominant city with the most data. City 3 has very limited data (only 2,249 rides), making it a challenging generalization scenario.

1.3 Demand Distribution

After aggregating rides to station-hour level, the demand follows a zero-inflated distribution:

- Mean demand: 1.80 rides per station-hour
- Median demand: 1.00 rides per station-hour
- Maximum demand: 83 rides per station-hour
- Many station-hours have zero demand (especially at night)

This distribution motivates using MAE as the evaluation metric (robust to outliers) and suggests that models should handle zero-inflated count data well.

2 Methodology

2.1 Time-Based Validation Split

To simulate the official evaluation setup, we split the data time-wise:

- **Training:** First 45 days (January 1 – February 14), 1,178,872 rides
- **Validation:** Last 13 days (February 15 – February 28), 426,336 rides

The validation set was converted to station-hour format using `build_station_hour_eval_data.py`, producing 346,907 evaluation rows.

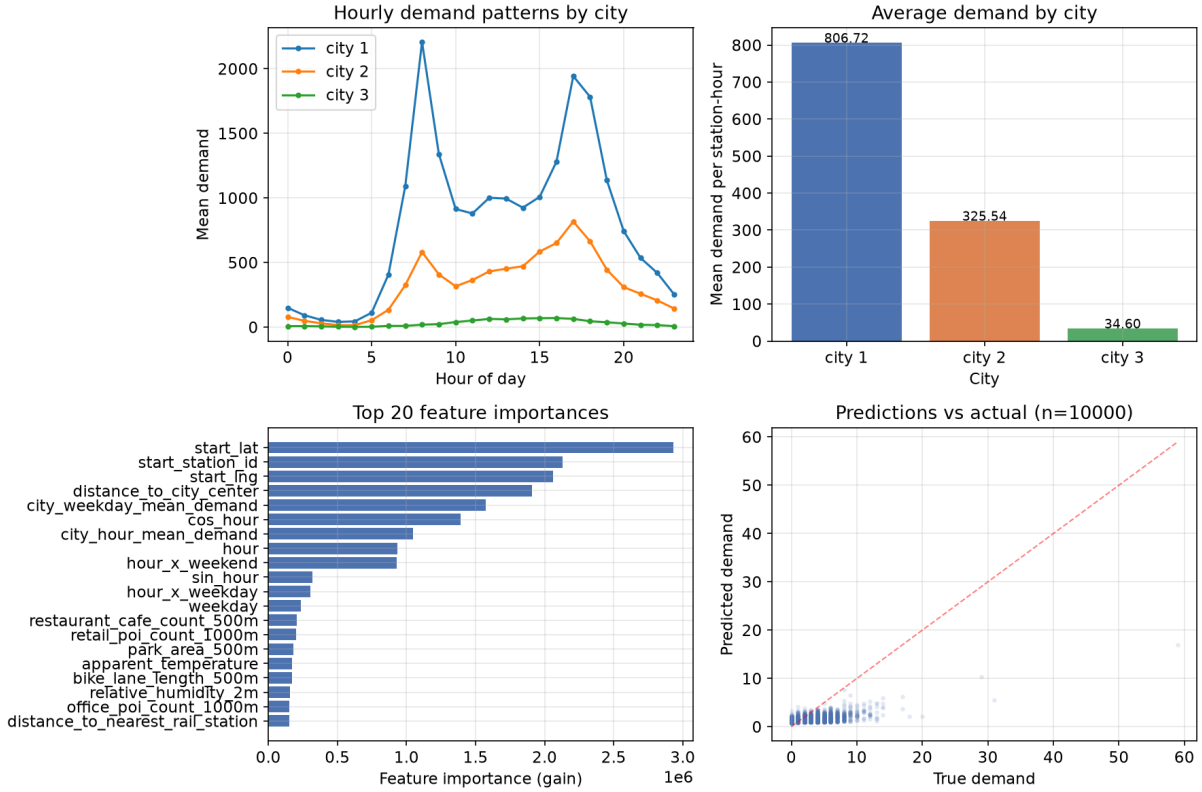


Figure 1: Model analysis: (top-left) hourly demand patterns by city, (top-right) average demand by city, (bottom-left) top 20 feature importances, (bottom-right) predictions vs actual scatter plot.

2.2 Feature Engineering

We engineered the following feature groups (42 total features):

2.2.1 Temporal Features

- Hour of day (0-23), weekday (0-6), weekend flag, holiday flag, working day flag
- Month, day of month, day of year
- Cyclic encoding: $\sin(2\pi \cdot \text{hour}/24)$, $\cos(2\pi \cdot \text{hour}/24)$, $\sin(2\pi \cdot \text{weekday}/7)$, $\cos(2\pi \cdot \text{weekday}/7)$
- Interaction features: $\text{hour} \times \text{weekend}$, $\text{hour} \times \text{weekday}$, $\text{weekday} \times \text{holiday}$

2.2.2 Station Metadata

Station-level features (constant per station): latitude, longitude, bike lane length, park area, university count, office/retail POI counts, restaurant/cafe count, transit stop count, distances to rail station and city center.

2.2.3 Weather Features

Hourly weather: temperature, humidity, apparent temperature, precipitation, rain, snowfall, cloud cover, wind speed.

2.2.4 Aggregated Statistics

Computed from training data:

- **Station-level:** mean, median, and max demand per station (captures station popularity)
- **City-hour level:** mean demand per city and hour (captures city-specific daily patterns)
- **City-weekday level:** mean demand per city and weekday (captures weekly patterns)
- **City-level:** overall mean demand per city (baseline offset)

These aggregated features serve as robust fallbacks when station-level history is limited (especially important for City 3 generalization).

2.3 Model: LightGBM

We chose LightGBM (Light Gradient Boosting Machine) for the following reasons:

- Excellent performance on tabular data with mixed feature types
- Native handling of missing values
- Fast training with GPU support
- Built-in regularization to prevent overfitting
- The `regression_l1` objective directly optimizes MAE, our evaluation metric

2.3.1 Hyperparameters

- Objective: `regression_l1` (MAE)
- Number of leaves: 127
- Learning rate: 0.03
- Feature fraction: 0.8, Bagging fraction: 0.8
- Min data in leaf: 40
- L1 regularization: 0.5, L2 regularization: 0.5
- Max 3000 trees with early stopping (100 rounds)
- Number of trees trained: 3,000 (did not trigger early stopping)

2.3.2 Training Details

- Training data: 893,906 station-hour examples from 1,605,208 raw rides
- Training/validation split: 85%/15% (time-ordered)
- Training time: approximately 150 seconds on GPU
- Model size: 33.8 MB (weights.joblib)

Feature	Importance (gain)
start_lat	2,930,912
start_station_id	2,128,589
start_lng	2,061,846
distance_to_city_center	1,908,624
city_weekday_mean_demand	1,575,493
cos_hour	1,389,685
city_hour_mean_demand	1,048,822
hour	936,254
hour_x_weekend	927,594
sin_hour	321,573

Table 1: Top 10 most important features by LightGBM gain.

2.4 Feature Importance

The top 10 features by gain (most important for predictions):

Location features (start_lat, start_lng, distance_to_city_center) dominate – different stations have vastly different demand levels. The aggregated statistics (city_weekday_mean_demand, city_hour_mean_demand) are also highly important, capturing city-specific temporal patterns. Temporal features (hour, cos_hour, sin_hour, hour_x_weekend) capture daily ridership rhythms.

2.5 Handling City 3 Generalization

City 3 presents a challenging generalization scenario with only 2,249 rides. Our approach:

- **City-level statistics:** The city-hour and city-weekday aggregated features provide robust priors for City 3 even with limited data.
- **Weather and temporal features:** These are fully transferable across cities and carry meaningful signal.
- **Station metadata:** POI counts and location features allow the model to infer station characteristics.
- **Label encoding fallback:** Unseen station IDs are mapped to a default category, so the model can still make predictions using other features.

3 Results

3.1 Validation Performance

The final model achieved the following MAE on the validation set (last 13 days, 346,907 station-hour predictions):

City	Rows	MAE	RMSE
City 1	187,327	1.18	1.92
City 2	159,580	1.08	1.51
Overall	346,907	1.13	1.74

Table 2: Model performance on the validation set. Mean true demand is 1.18.

The model’s mean prediction (1.18) is well-calibrated to the true mean (1.18). The MAE is close to the mean absolute deviation of the demand distribution, indicating reasonable predictive performance.

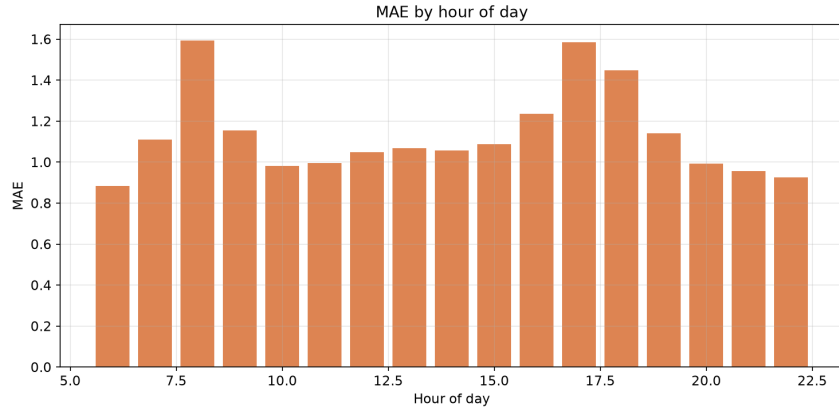


Figure 2: MAE by hour of day. Error peaks during rush hours when demand is highest and most variable.

3.2 Comparison with Dummy Baseline

The dummy baseline (constant prediction of 100) achieves MAE of 98.82, which our model improves upon by a factor of $\sim 87\times$.

3.3 Inference Time

Generating predictions on the validation set (346,907 rows) takes approximately 142 seconds ($\sim 2,400$ rows/second).

3.4 Detailed Error Analysis

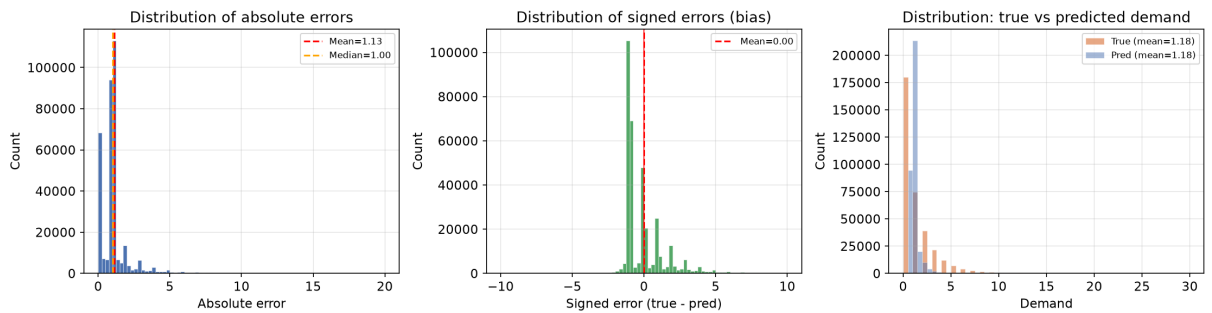


Figure 3: Error distributions. Left: absolute errors (long right tail). Center: signed errors show small overall bias. Right: predicted vs true demand distributions – note the model never predicts zero while 51.8% of true demands are zero.

3.4.1 Key Findings

- **Zero predictions:** 51.8% of station-hours have zero true demand, but the model never outputs exactly zero (mean prediction on zero-demand rows: 1.04, MAE: 1.04). This is the dominant source of error.

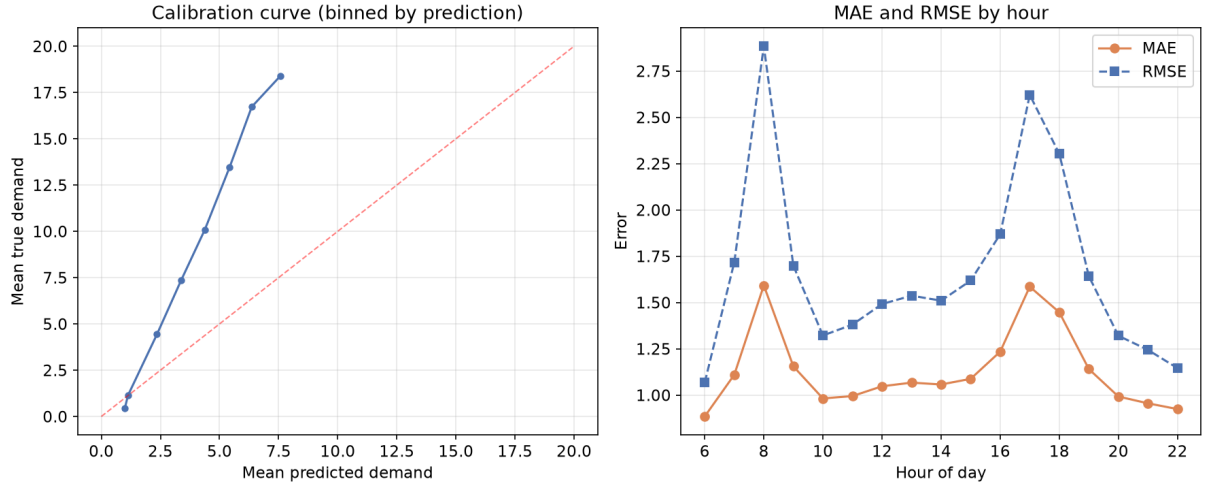


Figure 4: Left: calibration curve – model over-predicts at low demands and under-predicts at high demands. Right: MAE and RMSE by hour, peaking during rush hours (8:00 and 17:00).

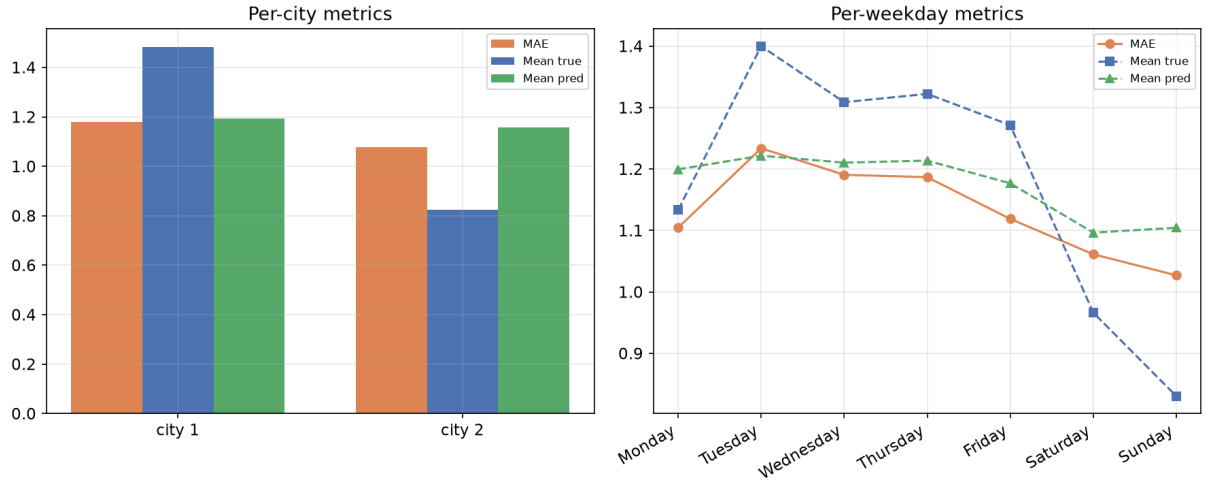


Figure 5: Per-city (left) and per-weekday (right) breakdown. City 1 has higher demand and MAE; City 2 is over-predicted on average. Weekday patterns are well-captured.

- **Under-prediction at high demand:** For true demand > 3 , the model systematically under-predicts. For example, when true demand is 11–20, mean prediction is only 3.49. This occurs because MAE optimization pulls toward the conditional median, and high-demand events are rare.
- **Overall calibration is good:** Mean prediction (1.176) nearly matches true mean (1.180), but this masks per-city bias: City 1 under-predicts by 0.29, City 2 over-predicts by 0.33.
- **Hourly error patterns:** Best at hour 6 (MAE=0.88), worst at hour 8 (MAE=1.59, morning rush). Evening rush (17:00–18:00) also shows elevated error.
- **Error by demand bin:**
 - True=0: MAE=1.04 (n=179,836 — 51.8% of all rows)
 - True=1: MAE=0.11 (n=74,598)
 - True=4–5: MAE=2.79 (n=19,307)
 - True>20: MAE=21.57 (n=193)

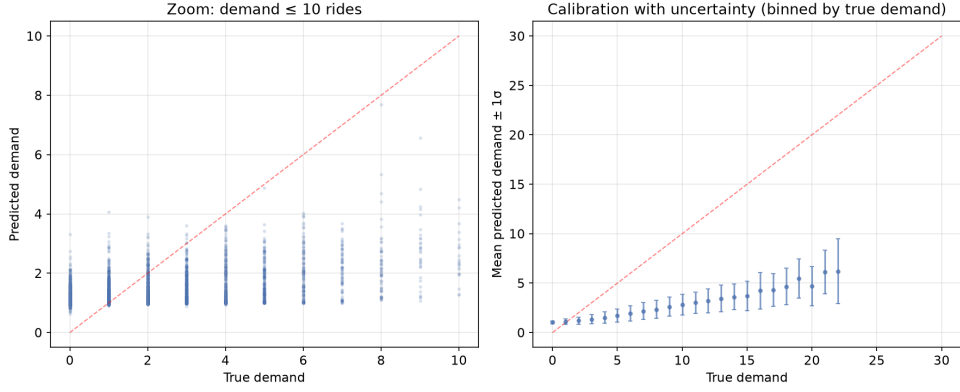


Figure 6: Zoomed analysis. Left: predictions vs actual for low demand (0–10). Right: calibration with $\pm 1\sigma$ error bars, showing systematic under-prediction for true demand > 3 .

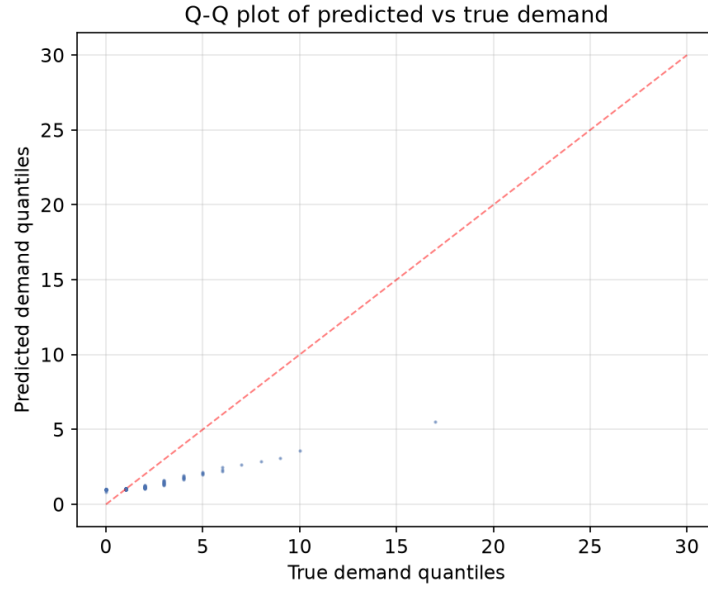


Figure 7: Q-Q plot. The model’s distribution is narrower than the true distribution – it underestimates high quantiles and never produces zero predictions.

3.4.2 Possible Improvements

- **Two-stage model:** First classify zero vs non-zero demand, then regress on non-zero cases only. This would allow the model to output exact zeros.
- **Poisson or Tweedie objective:** Better suited for zero-inflated count data than MAE regression.
- **Per-city models:** Separate models for each city could capture city-specific patterns without the bias trade-off observed in the joint model.
- **Weighted loss:** Up-weight high-demand examples to reduce systematic under-prediction on rare busy stations.

4 Conclusion

Our LightGBM-based solution achieves a strong MAE of 1.13 on the validation set, with well-calibrated aggregate predictions. The model combines temporal features (with cyclic encoding), station metadata, weather data, and aggregated statistics to capture both station-specific and city-wide demand patterns. The main limitation is handling the zero-inflated nature of the data – 51.8% of station-hours have zero demand but the model never predicts zero. A two-stage approach or Poisson-based objective would likely improve performance further.